

COLLARDRIVE · REQUIREMENTS

CollarDrive requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/collardrive		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-03	Amish Chadha	TRL 3 status from CLD-CAL-001 v0.2 on the constructable design (CLD-DDR-002); requirements decided under Amish's 2026-10-03 pre-approval
0.2	2026-10-03	Amish Chadha	TRL 2 update; targets confirmed as planning figures; R10 restated against the value-engineering target; concept status
0.1	2026-09-30	Amish Chadha	Initial scaffold

CollarDrive requirements

All ten requirements are met on paper or by design on the constructable design (CLD-CAL-001 v0.2, Table 4). None is missed. One has a thin margin: at the 30 m design depth the two pump belts carry 92 % of their allowable pull, so the 40 mm rising main is not used below 30 m. The targets are planning figures, decided by Amish under his pre-approval of 2026-10-03 (CLD-DDR-001); they are to be checked with the co-design partner and by test at TRL 4.

Table 1. Requirements and status at TRL 3.

ID	Requirement	Target	Verification (TRL 3 or later)	Status (CLD-CAL-001)
R1	Engine stays on the surface	No part with combustion below collar level	Design review of the general arrangement	Met by design: engine on the drive module, 1.9 m or more from the shaft centre
R2	Exhaust kept away from the shaft	Exhaust outlet at least 6 m (20 ft) from the collar and the blower intake	Layout check; later a carbon monoxide reading at the blower intake during a run	Met on paper: 7.05 m from the edge of a 1.5 m collar and 6.46 m from the intake, outlet 2.4 m high
R3	Dewatering flow	At least 60 L/min at 20 m head	Calculation; later a flow test on a test well or tower	Met on paper: 75.7 L/min
R4	Dewatering depth	Useful flow from 60 m (200 ft)	Calculation; later a flow test at depth	Met on paper with the 25 mm main below 45 m: 19.3 L/min
R5	Air delivery	At least 0.1 m ³ /s (about 210 cfm) at the end of 30 m of duct	Calculation; later an anemometer at the duct outlet	Met on paper: 0.189 m ³ /s (0.153 m ³ /s at the end of 60 m)
R6	Shared drive	Pump and blower run together from one engine of 5 kW (about 7 hp) or less	Power calculation; later a bench run	Met on paper: 0.78 kW at 30 m from a 4.8 kW class engine
R7	Fits real collars	Frame spans collars from 0.8 to 1.5 m across	Model; later a fit check at partner sites	Met by design: sleeper edges at least 0.35 m clear of a 1.5 m collar
R8	Locally buildable	Built from common steel sections with a stick welder and hand tools	Build plan review with a local workshop	Met by design: RHS, SHS, plate and bar; heaviest piece 69.6 kg; shafts turned and keyed by a local machine shop
R9	Guarded drive	All belts, pulleys and shafts fully guarded; the wheel cannot run back	Guarding review; later an inspection	Met by design: belt guard, layshaft cover, wheel hood, meshed blower inlet, ratchet backstop
R10	Cost against the value-engineering target	Value-engineering target USD 3,500 for the full kit	Costed bill of materials	Under the target: estimated USD 2,535, USD 965 under

■ Assumptions

- Miners already own or can buy a small petrol or diesel engine of the 4.8 kW (6.5 hp) class.
- A rope pump handles the grit in shaft water with an acceptable rope and piston life; to be checked at TRL 4.

- The planning figure of 0.05 m³/s of fresh air per person at the face is a starting point, not a regulation; the co-design partner and local rules may require more.
- Shaft collars are stable enough for sleepers set back 0.35 m from their edge.

■ Safety

Safety: Meeting these requirements keeps the engine and its exhaust out of the shaft; it does not make the shaft safe to enter. A gas detector (carbon monoxide and oxygen) is used before and during any work below ground. The collar is fenced and the frame is never stood on. All moving parts are guarded, and the engine is stopped before anyone touches the drive.